
EMCH 574 – Vibration and Waves Homework 01

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Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#).
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) [reference](#).
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the *y*-axis scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 Pendulum

A.1 Analyze Pendulum

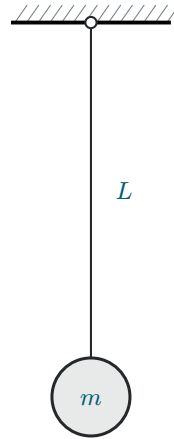


Figure 1: Pendulum of length L and mass m .

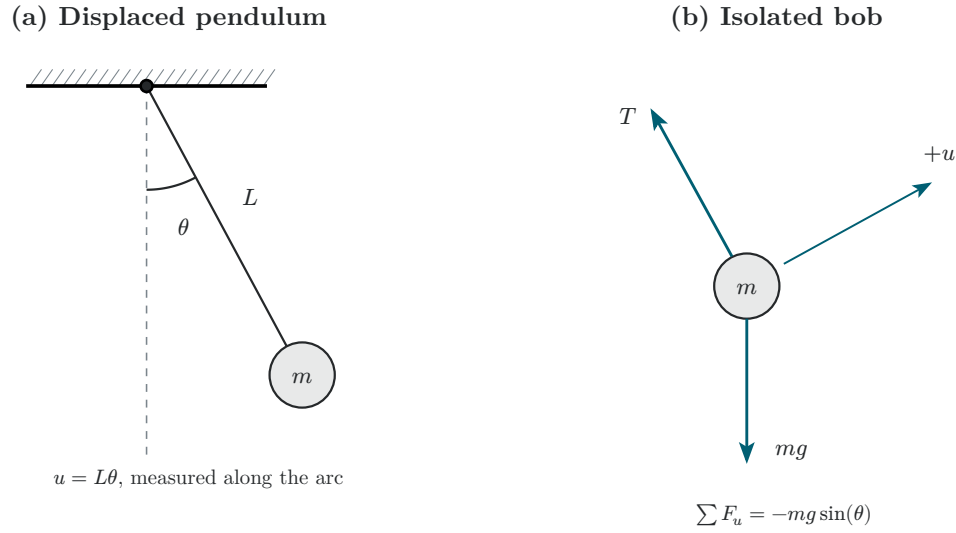
Consider a pendulum of length L and mass m as shown in Figure 1.

Do the following.

- (1) Draw the pendulum displaced by an angle θ from the vertical.
- (2) Draw free-body diagram (FBD).
- (3) Write Newton's law of motion (NLM) equation.
- (4) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (5) Deduce the governing equation.
- (6) Assume $\theta(t) = e^{st}$ and substitute into the governing equation.
- (7) Deduce and solve the characteristic equation.
- (8) What type are the roots $s_{1,2}$? Choose one of the following options:
 - (a) Real.
 - (b) Imaginary.
 - (c) Complex.
- (9) Define the natural frequency ω_n .
- (10) Express the solutions s_1 and s_2 of the characteristic equation in terms of ω_n .
- (11) Write the solution in terms of exponential functions of $\omega_n t$.
- (12) Write the solution in terms of trigonometric functions of $\omega_n t$.
- (13) Write the Hz frequency f_n .

- (14) Write the period of oscillation τ .
- (15) Write the particle motion $u(t)$ in terms of:
- (a) Exponential functions.
 - (b) Trigonometric functions.
- (16) Solve the initial value problem (IVP) and find the response $u(t)$ for:
- (i) Initial displacement u_0 .
 - (ii) Initial velocity $\dot{u}_0$.
 - (iii) Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to A.1

**Figure 2:** Free-body diagram for Problem A.1.

Take positive θ in the direction shown and let $u = L\theta$ be the tangential arc displacement. Then $\ddot{u} = L\ddot{\theta}$. Assume a point mass, a massless inextensible rod or taut string of length L , a fixed frictionless pivot, uniform gravity, planar motion, and negligible air resistance.

The weight is mg , and the tension T has no tangential component. Therefore, the tangential force is the restoring component of gravity,

$$\sum F_u = -mg \sin \theta = m\ddot{u} = mL\ddot{\theta} \quad (\text{A.1.3})$$

For small angular displacement in radians, $\sin \theta \approx \theta$. Substituting this approximation gives the equation of motion

$$mL\ddot{\theta} + mg\theta = 0 \quad (\text{A.1.4})$$

Dividing by mL gives the governing equation

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \quad (\text{A.1.5})$$

To solve the governing equation, assume $\theta = e^{st}$. Then $\ddot{\theta} = s^2 e^{st}$, so substitution gives

$$\left(s^2 + \frac{g}{L}\right) e^{st} = 0.$$

Because $e^{st} \neq 0$, the characteristic equation is

$$\boxed{s^2 + \frac{g}{L} = 0} \quad (\text{A.1.7})$$

Since $g/L > 0$, solving this equation gives purely imaginary roots. Therefore, the answer to item 8 is

$$\boxed{\text{Imaginary}} \quad (\text{A.1.8.b})$$

Define the natural angular frequency by $\omega_n^2 = g/L$. The natural frequency is therefore

$$\boxed{\omega_n = \sqrt{\frac{g}{L}}} \quad (\text{A.1.9})$$

Using this definition, the roots become

$$\boxed{s_1 = i\omega_n, \quad s_2 = -i\omega_n} \quad (\text{A.1.10})$$

The two roots produce the exponential angular response

$$\boxed{\theta(t) = D_1 e^{i\omega_n t} + D_2 e^{-i\omega_n t}} \quad (\text{A.1.11})$$

Using Euler's identity, the same response can be written as

$$\boxed{\theta(t) = A_\theta \cos(\omega_n t) + B_\theta \sin(\omega_n t)} \quad (\text{A.1.12})$$

Because $\omega_n = 2\pi f_n$ and $f_n = 1/\tau$, the frequency and period are

$$\boxed{f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}} \quad (\text{A.1.13})$$

$$\boxed{\tau = \frac{1}{f_n} = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{L}{g}}} \quad (\text{A.1.14})$$

Multiplying the angular response by L gives the particle-motion response. In exponential form, this is

$$\boxed{u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t}} \quad (\text{A.1.15.a})$$

Euler's identity also gives the trigonometric forms

$$\boxed{u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) = C \sin(\omega_n t + \varphi)} \quad (\text{A.1.15.b})$$

For real A and B , the amplitude and phase are $C = \sqrt{A^2 + B^2}$ and $\varphi = \text{atan2}(A, B)$.

Differentiate the trigonometric response and evaluate it at $t = 0$. The initial conditions give

$$u(0) = A = u_0, \quad \dot{u}(0) = \omega_n B = v_0, \quad v_0 = \dot{u}_0.$$

With initial displacement only, $v_0 = 0$, so

$$\boxed{u(t) = u_0 \cos(\omega_n t)} \quad (\text{A.1.16.i})$$

With initial velocity only, $u_0 = 0$, so

$$\boxed{u(t) = \frac{v_0}{\omega_n} \sin(\omega_n t)} \quad (\text{A.1.16.ii})$$

With both initial conditions applied, the response is

$$\boxed{u(t) = u_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t)} \quad (\text{A.1.16.iii})$$

The corresponding exponential constants are

$$C_1 = \frac{1}{2} \left(u_0 - \frac{iv_0}{\omega_n} \right), \quad C_2 = \frac{1}{2} \left(u_0 + \frac{iv_0}{\omega_n} \right).$$

The mass cancels when the governing equation is divided by m , so the ideal small-angle pendulum frequency depends on g and L , but not on m .

A.2 Calculate Pendulum Response to Initial Displacement

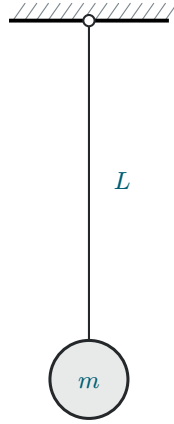


Figure 3: Pendulum of length L and mass m .

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L_{\text{nominal}} = 565 \text{ mm}$ and mass $m = 52 \text{ kg}$ as shown in Figure 3. When manufactured in series production, the pendulum lengths are not exact because of manufacturing tolerances. The lengths of the ten pendulum sets are

$$L = [561.535, 562.691, 563.845, 563.845, 565.000, \\ 565.000, 566.155, 566.155, 567.309, 568.464] \text{ mm}.$$

Do the following.

- (1) For the ten pendulum sets, calculate.
 - (a) Natural frequency ω_n in rad/s.
 - (b) Hz natural frequency f_n .
 - (c) Period of oscillation τ .
 - (d) Display in table format L, ω_n, f_n, τ .
- (2) Mean values: calculate and display the mean values of ω_n, f_n, τ .
- (3) Initial value problem (IVP). Assume initial displacement $u_0 = 2 \text{ mm}$ and zero initial velocity. Use the mean values of ω_n, f_n , and τ to complete items (a)–(c).
 - (a) Calculate and display the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
 - (b) Calculate and display the duration $t_{N_{\text{osc}}}$ of N_{osc} oscillations and the corresponding position of the mass $u_{N_{\text{osc}}}$ for $N_{\text{osc}} = 10$.
 - (c) Plot the response of the system for N_{osc} oscillations and use a data tip to find $u(t_1)$ at t_1 .

Solution to A.2

Use SI units by converting each length from millimetres to metres. For set j , the pendulum relation from A.1 gives

$$\omega_{n,j} = \sqrt{\frac{g}{L_j}} \quad (\text{A.2.1.a})$$

The frequency and period follow from $\omega_{n,j} = 2\pi f_{n,j}$ and $f_{n,j} = 1/\tau_j$, so

$$f_{n,j} = \frac{\omega_{n,j}}{2\pi} \quad (\text{A.2.1.b})$$

$$\tau_j = \frac{2\pi}{\omega_{n,j}} \quad (\text{A.2.1.c})$$

Using $g = 9.81 \text{ m/s}^2$ and the ten lengths gives the following values for A.2.1.d.

Set	L (mm)	ω_n (rad/s)	f_n (Hz)	τ (s)
1	561.535	4.18	0.665	1.503
2	562.691	4.18	0.665	1.505
3	563.845	4.17	0.664	1.506
4	563.845	4.17	0.664	1.506
5	565.000	4.17	0.663	1.508
6	565.000	4.17	0.663	1.508
7	566.155	4.16	0.663	1.509
8	566.155	4.16	0.663	1.509
9	567.309	4.16	0.662	1.511
10	568.464	4.15	0.661	1.513

Compute the mean of each calculated column using $\bar{q} = \frac{1}{10} \sum_{j=1}^{10} q_j$. The mean values required in A.2.2 are

$$\overline{\omega_n} = 4.17 \text{ rad/s} \quad (\text{A.2.2})$$

$$\overline{f_n} = 0.663 \text{ Hz} \quad (\text{A.2.2})$$

$$\overline{\tau} = 1.508 \text{ s} \quad (\text{A.2.2})$$

For A.2.3, the initial velocity is zero, so the A.1 response reduces to

$$u(t) = u_0 \cos(\overline{\omega_n} t) = 2 \cos(4.166893t) \text{ mm} \quad (\text{A.2.3})$$

At $t_1 = 10$ s, substitution into the response gives

$$u(t_1) = 2 \cos(4.166893 \times 10) \text{ mm} = -1.352 \text{ mm} \quad (\text{A.2.3.a})$$

For ten oscillations, multiply the mean period by $N_{\text{osc}} = 10$. The duration and corresponding position are

$$t_{N_{\text{osc}}} = 10\bar{\tau} = 15.08 \text{ s} \quad (\text{A.2.3.b})$$

$$u_{N_{\text{osc}}} = u(10\bar{\tau}) = 2.000 \text{ mm} \quad (\text{A.2.3.b})$$

Plot the response over $0 \leq t \leq 10\bar{\tau}$ using the boxed A.2.3 expression. The figure below shows the ten-oscillation response and marks the value at $t_1 = 10$ s. The plotted value is

$$u(10) = -1.352 \text{ mm} \quad (\text{A.2.3.c})$$

A2. Pendulum released from 2 mm

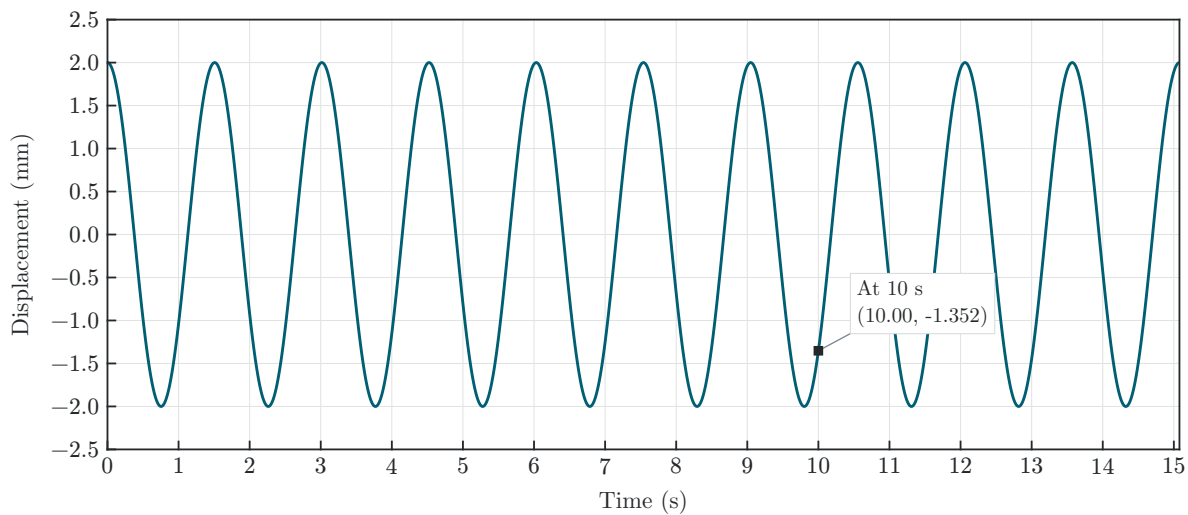


Figure 4: A.2.3.c. Pendulum response over ten oscillations with a data tip at $t = 10$ s.

MATLAB code. *Script used to reproduce Figure 4.*

```
01 u0 = 2;  
02 wn = 4.166893;  
03 t1 = 10;  
04 Nosc = 10;  
05  
06 tau = 2*pi/wn;  
07 t = linspace(0,Nosc*tau,1000);  
08 u = u0*cos(wn*t);  
09  
10 plot(t,u,t1,u0*cos(wn*t1),'ks');  
11 xlabel('Time (s)');  
12 ylabel('u (mm)');
```

A.3 Calculate Pendulum Response to Impact

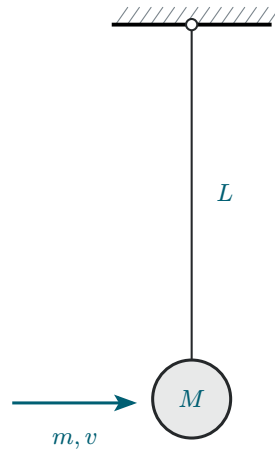


Figure 5: *Impact excitation of a pendulum system.*

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L = 565 \text{ mm}$ and mass $M = 52 \text{ kg}$ as shown in Figure 5. The pendulum is initially at rest hanging vertically. A small projectile of mass $m = 22 \text{ g}$ and velocity $v = 945 \text{ m/s}$ hits the mass M and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/sec natural frequency ω_n and period of oscillation τ of the pendulum with projectile embedded into it.
- (3) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (4) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (5) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (4).
- (6) Discuss your results.

Solution to A.3

The input values required for A.3.1 are shown below. Convert them to SI units before calculating the response. The projectile mass is denoted by m_p so that it is distinct from the pendulum mass M .

Quantity	Value	Unit
g	9.81	m/s ²
L	0.565	m
M	52.0	kg
m_p	0.0220	kg
v_p	945	m/s
t_1	10.0	s
N_{osc}	10	1

For the pendulum, the small-angle restoring motion gives the undamped equation

$$\ddot{u} + \frac{g}{L}u = 0.$$

Therefore, the natural frequency does not depend on the total mass, because the mass cancels from the ideal pendulum equation. The natural frequency and period are

$$\omega_n = \sqrt{\frac{g}{L}} = 4.16687 \text{ rad/s} \quad (\text{A.3.2})$$

$$\tau = \frac{2\pi}{\omega_n} = 1.50789 \text{ s} \quad (\text{A.3.2})$$

During the short impact, the projectile embeds in the pendulum mass, so linear momentum is conserved. Since the pendulum mass starts at rest, the common velocity immediately after impact satisfies

$$m_p v_p + M(0) = (M + m_p) \dot{u}_0.$$

Solving for the initial velocity gives

$$\dot{u}_0 = \frac{m_p v_p}{M + m_p} = 0.399639 \text{ m/s} \quad (\text{A.3.3})$$

The pendulum is still vertical immediately after impact, so $u_0 = 0$. Substitution into the undamped initial-value solution gives

$$u(t) = u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t).$$

Because $u_0 = 0$, the response becomes

$$u(t) = \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) = 95.9085 \sin(4.16687t) \text{ mm} \quad (\text{A.3.4})$$

At $t_1 = 10 \text{ s}$, substitution into the response gives

$$u(t_1) = 95.9085 \sin(4.16687 \times 10) \text{ mm} = -70.6461 \text{ mm} \quad (\text{A.3.4})$$

Ten oscillations span the interval

$$t_{\text{end}} = N_{\text{osc}} \tau = 10(1.50789) = 15.0789 \text{ s} \quad (\text{A.3.5})$$

The response reaches its first positive maximum one quarter period after impact. The largest absolute response is therefore

$$|u|_{\text{max}} = \frac{\dot{u}_0}{\omega_n} = 95.9085 \text{ mm} \quad (\text{A.3.5.a})$$

The ten-oscillation plot includes data tips at the first maximum and at $t_1 = 10 \text{ s}$. The plotted readings are

$$u_{\text{max}} = 95.9 \text{ mm} \quad (\text{A.3.5.a})$$

$$u(10 \text{ s}) = -70.6 \text{ mm} \quad (\text{A.3.5.b})$$

A3. Response after an embedded-projectile impact

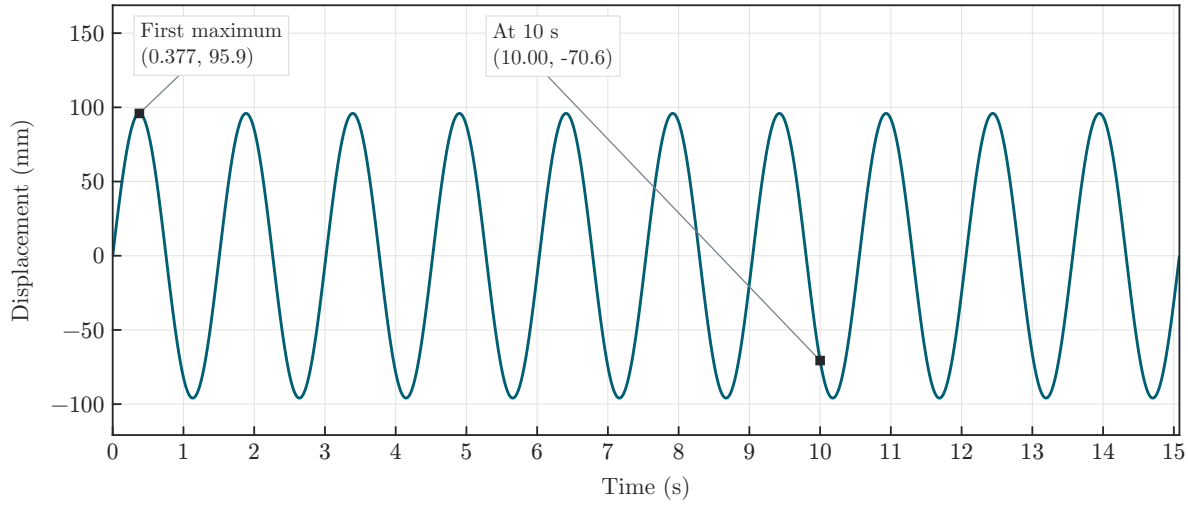


Figure 6: A.3.5.a-b. Impact response with data tips.

The projectile transfers its momentum to the combined mass, which produces a nonzero initial velocity but no initial displacement. The response consequently starts at zero, reaches its first maximum at $t = \tau/4 = 0.377$ s, and then repeats with constant amplitude because damping is neglected. The maximum displacement corresponds to an angular amplitude of approximately $u_{\max}/L = 0.170$ rad = 9.73° , so the small-angle model is reasonable for this calculation. At 10 s, the pendulum is moving through the negative side of the motion rather than being at a maximum, which gives the plotted value of -70.6 mm.

MATLAB code. *Script used to plot the ten-oscillation response.*

```
01 g = 9.81;
02 L = 0.565;
03 M = 52;
04 mp = 0.022;
05 vp = 945;
06 t1 = 10;
07 Nosc = 10;
08 u0 = 0;
09
10 wn = sqrt(g/L);
11 tau = 2*pi/wn;
12 udot0 = mp*vp/(M+mp);
13 A = udot0/wn;
14
15 t = linspace(0,Nosc*tau,1000);
16 u = u0*cos(wn*t) + A*sin(wn*t);
17 u1 = u0*cos(wn*t1) + A*sin(wn*t1);
18
19 plot(t,1000*u,tau/4,1000*A,'ks',t1,1000*u1,'ks');
20 xlabel('Time (s)');
21 ylabel('u (mm)');
22 grid on;
```

2 Spring-Mass-Damper Vibrating Systems

B.1 Analyze Spring-Mass Vibrating System

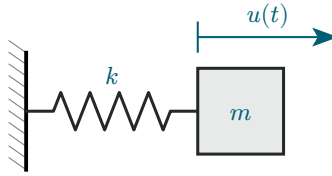


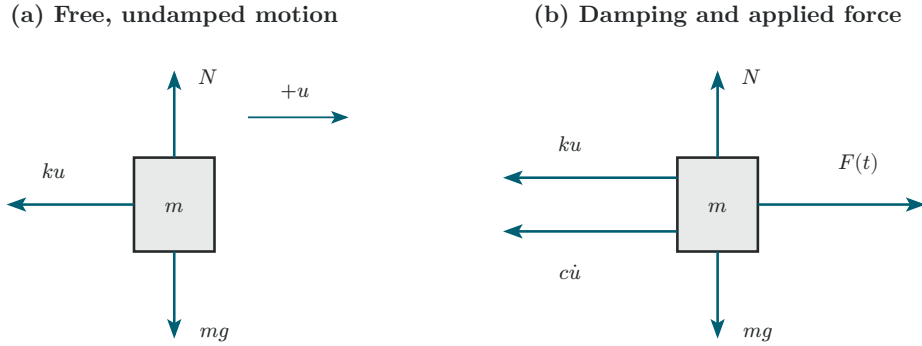
Figure 7: Spring-mass vibrating system.

Consider a spring-mass vibrating system as shown in Figure 7.

Do the following.

- (1) Draw free-body diagram (FBD).
- (2) Write NLM equation.
- (3) Deduce the equation of motion (EOM).
- (4) Deduce the governing equation. Clearly state your assumptions.
- (5) Assume $u(t) = e^{st}$ and substitute into the governing equation.
- (6) Deduce the characteristic equation.
- (7) Define the natural frequency ω_n in terms of m, k .
- (8) Solve the characteristic equation and find $s_{1,2}$ in terms of ω_n .
- (9) What type are the roots $s_{1,2}$? Choose one of the following options:
 - (a) Real.
 - (b) Imaginary.
 - (c) Complex.
- (10) Write the solution in terms of exponential functions.
- (11) Write the solution in terms of trigonometric functions.
- (12) Solve the initial value problem (IVP) and find the response $u(t)$ for:
 - (i) Initial displacement u_0 .
 - (ii) Initial velocity $\dot{u}_0$.
 - (iii) Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to B.1



Set $c = 0$ for an undamped system; set $F = 0$ for free vibration.

For beam models, N represents the balanced vertical support reaction.

Figure 8: B.1.1. Free-body diagram for the spring-mass system.

Take displacement $u(t)$ as positive to the right and measure it from the equilibrium position. Assume one degree of freedom, a point mass m , a massless linear spring with stiffness k , frictionless horizontal motion, and no damping or external force. Because the spring force opposes the displacement, its horizontal force is

$$\sum F_x = -ku(t)$$

Applying Newton's second law to the mass gives

$$\sum F_x = m\ddot{u}(t) \quad (\text{B.1.2})$$

Equating the spring force and the inertial force produces the equation of motion,

$$m\ddot{u}(t) + ku(t) = 0 \quad (\text{B.1.3})$$

Since m is nonzero, divide by m to obtain the governing equation in normalized form,

$$\ddot{u}(t) + \frac{k}{m}u(t) = 0 \quad (\text{B.1.4})$$

Assume an exponential response $u(t) = e^{st}$. Its derivatives are $\dot{u}(t) = se^{st}$ and $\ddot{u}(t) = s^2e^{st}$. Substitution into the equation of motion gives

$$(ms^2 + k)e^{st} = 0 \quad (\text{B.1.5})$$

Because e^{st} is never zero, the factor multiplying it must vanish. Therefore, the characteristic equation is

$$ms^2 + k = 0 \quad (\text{B.1.6})$$

Define the undamped natural frequency by

$$\omega_n = \sqrt{\frac{k}{m}} \quad (\text{B.1.7})$$

The characteristic equation then becomes $s^2 + \omega_n^2 = 0$, so its two roots are

$$s_{1,2} = \pm i\omega_n \quad (\text{B.1.8})$$

For positive m and k , these roots have zero real part and nonzero imaginary parts. Thus, the roots are

$$\text{Imaginary} \quad (\text{B.1.9})$$

Using the two distinct roots in the exponential solution gives

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} \quad (\text{B.1.10})$$

Applying Euler's identity converts this response into the equivalent real trigonometric form,

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) \quad (\text{B.1.11})$$

Let $u_0 = u(0)$ and $\dot{u}_0 = \dot{u}(0)$. Evaluating the trigonometric solution at $t = 0$ gives $A = u_0$. Differentiating and evaluating at $t = 0$ gives $\dot{u}_0 = \omega_n B$, so $B = \dot{u}_0/\omega_n$. Therefore, the three requested initial-condition cases are

$$\begin{aligned} u(t) &= u_0 \cos(\omega_n t) & (\text{B.1.12.i}) \\ u(t) &= \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) & (\text{B.1.12.ii}) \\ u(t) &= u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) & (\text{B.1.12.iii}) \end{aligned}$$

B.2 Analyze Spring-Mass-Damper Vibrating System

Consider the 1-dof damped system consisting of a spring, k , mass, m , and viscous damper, c , as presented in Figure 9. Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.

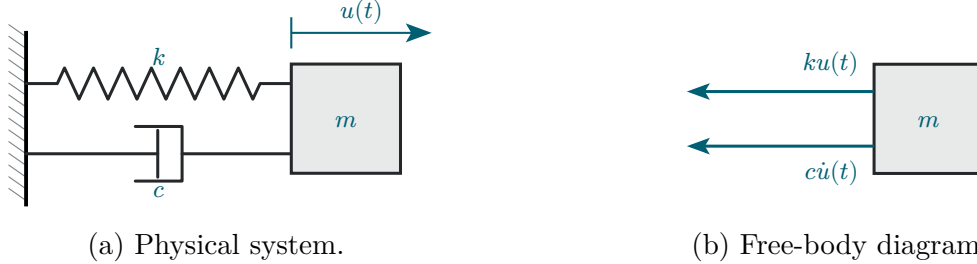


Figure 9: Mass-spring-damper vibrating system.

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k .
- (5) Assume $u(t) = e^{st}$ and substitute into EOM.
- (6) Deduce the characteristic equation.
- (7) Find the roots $s_{1,2}$ of the characteristic equation.
- (8) Define critical damping c_{cr} in terms of m and k .
- (9) Define damping ratio ζ .
- (10) Discuss the values of ζ for the three damping cases:
 - (a) Underdamped.
 - (b) Critically damped.
 - (c) Overdamped.
- (11) Express the physical damping c in terms of damping ratio ζ , mass m , and stiffness k .
- (12) Define the natural frequency ω_n in terms of m and k .
- (13) Express in terms of ω_n , ζ the $s_{1,2}$ roots found at (7).
- (14) Assume underdamped system and express the $s_{1,2}$ roots of (13) as complex numbers in terms of ω_n and ζ .
- (15) Define the damped frequency ω_d .
- (16) Substitute (15) into (14) and express the roots s_1 and s_2 in terms of ω_n , ω_d , ζ .
- (17) Assume underdamped conditions and state the inequality that ζ should satisfy.

- (18) Write the general solution of the ODE deduced at (4) in terms of constants C_1 and C_2 multiplying exponential functions of s_1 and s_2 .
- (19) Use (16) to write the solution (18) in terms of exponential functions of ω_n , ω_d , ζ .
- (20) Rewrite the solution (19) in terms of an exponential decay function multiplying the sum of two complex exponentials.
- (21) Rewrite the solution (20) in terms of an exponential decay function multiplying a trigonometric function and some constants C and ϕ to be determined from the initial conditions.
- (22) Sketch the functions discussed at (21).
- (23) Solve the initial value problem (IVP) for $\zeta < 1$ and find the response $u(t)$ for initial displacement u_0 with zero initial velocity.
- (24) Solve the initial value problem (IVP) and find the response $u(t)$ for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.

B.3 Calculate Spring-Mass-Damper Response to Initial Displacement

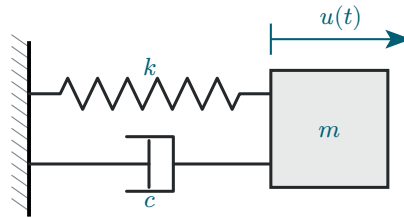


Figure 10: *Spring-mass-damper vibrating system.*

Consider a spring-mass vibrating system with $m = 12 \text{ kg}$, $k = 655 \text{ N/m}$, $c = 19 \text{ N/(m/s)}$ as shown in Figure 10. The vibrating system is given an initial displacement $u_0 = 3.5 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (3) Calculate the critical damping c_{cr} .
- (4) Calculate the damping ratio ζ .
- (5) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (6) Calculate the damped period of oscillation τ_d .
- (7) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2 .
- (8) Calculate the constants A , B and then the constants C , ϕ .
- (9) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (10) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (11) Plot the response of the system for $N_{osc} = 10$ oscillations and use datatips on the plot to find the following:
 - (a) The largest response.
 - (b) The response value calculated at item (9).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (10).
- (12) Discuss your results.

B.4 Calculate Truck Suspension Response

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a k - m - c spring-mass-damper system as shown in Figure 11.

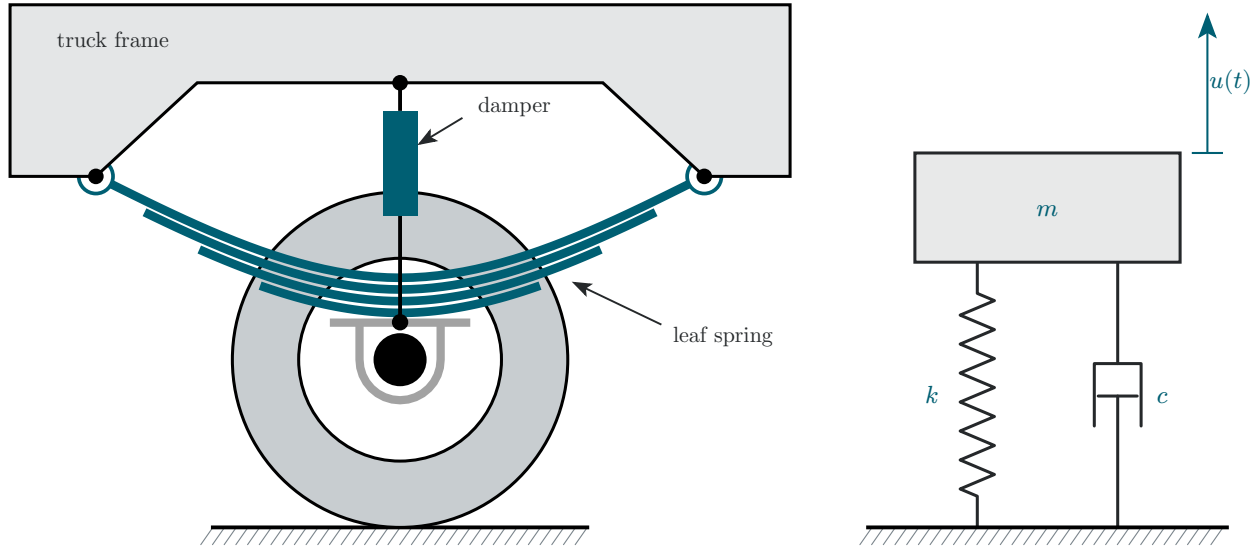


Figure 11: Truck suspension consisting of a leaf spring and an oil damper. The wheel contact is treated as the road datum in the one-degree-of-freedom idealization.

The leaf spring can be approximated by a uniform beam of length $L = 830 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 65 \text{ mm}$, made of steel with $E = 200 \text{ GPa}$. The truck mass apportioned to the wheel can be estimated as $m = 457 \text{ kg}$. Assume initial displacement $u_0 = 125 \text{ mm}$ and zero initial velocity $\dot{u}_0 = 0$.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I of the beam cross section.
- (3) Calculate the flexural stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (6) Calculate the critical damping c_{cr} .
- (7) Assume a 25% damping ratio, i.e., $\zeta = 0.25$, and calculate the physical damping c . Print ζ and c .
- (8) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (9) Calculate the damped period of oscillation τ_d .
- (10) Calculate the characteristic-equation roots s_1, s_2 and the constants C_1, C_2 .
- (11) Calculate the constants A, B and then the constants C, ϕ .

-
- (12) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (13) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (14) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and use datatips on the plot to find the following:
- (a) The largest response.
 - (b) The response value calculated at item (12).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (13).
- (15) Discuss your results.
- (16) Graduate work. Assume the damper is worn out such that ζ is reduced to $\zeta = 10\%$.
- (a) Repeat the calculation of items (6) through (12).
 - (b) Compare and discuss the two results.

B.5 Overdamped Spring-Mass-Damper Systems

Consider a spring-mass-damper system with natural frequency $f_n = 6.25 \text{ Hz}$ and various values of the damping ratio ζ . The system is given an initial displacement $u_0 = 13 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/s frequency ω_n .
- (3) Assume overdamped conditions $\zeta > 1$ and develop the solution as follows:
 - (a) Write the expressions of the roots s_1, s_2 in terms of ω_n and ζ .
 - (b) Write the solution $u(t)$ in terms of s_1, s_2 and constants C_1, C_2 .
 - (c) Write the expression of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Take $\zeta = 2.3$ and calculate the numerical values of s_1, s_2, C_1 , and C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3 \text{ s}$.
- (4) Assume critically damped conditions $\zeta = 1$ and complete items (a)–(e).
 - (a) Write the expressions of the two partial solutions $u_1(t), u_2(t)$ in terms of ω_n .
 - (b) Write the expression of the complete solution in terms of ω_n and C_1, C_2 .
 - (c) Write the expressions of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Calculate the numerical values of C_1, C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3 \text{ s}$.
- (5) Graduate work. Derive the ODE for $\zeta = 1$ and verify that the two solutions $u_1(t)$ and $u_2(t)$ from (4)(a) satisfy the ODE.
- (6) Graduate work. Some race-car manufacturers design for critical damping, whereas others use an overdamped design. Discuss the advantages and disadvantages of the two options.

B.6 Analyze Spring-Mass-Damper Forced Vibration

Consider a mass-spring-damper vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 12.

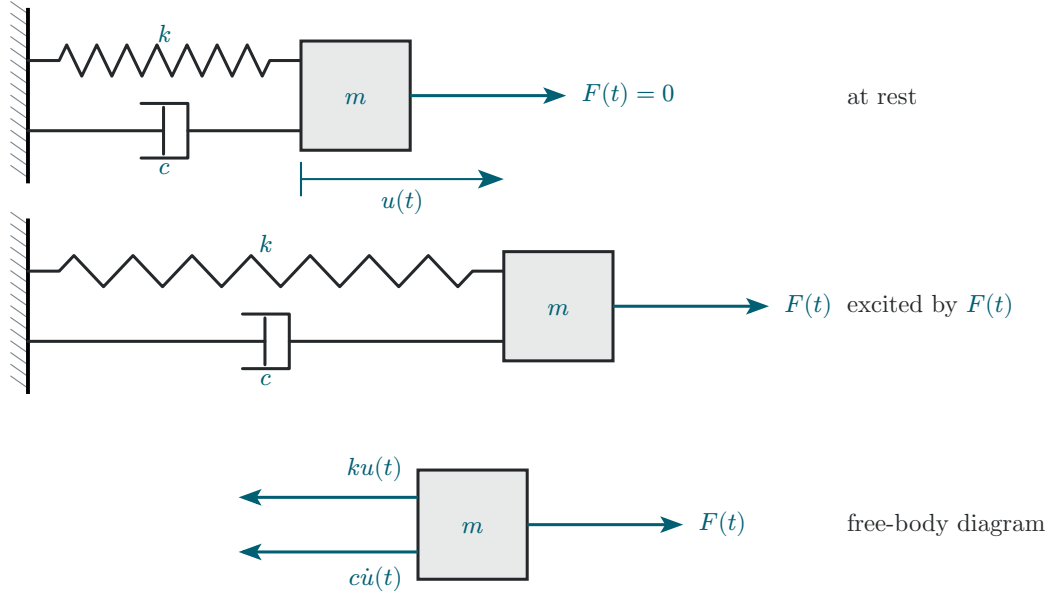


Figure 12: Spring-mass-damper system excited by force $F(t)$.

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k , $F(t)$.
- (5) Assume harmonic excitation $F(t) = \hat{F} e^{i\omega t}$, substitute in (4), and divide by m .
- (6) Define the normalized excitation amplitude $\hat{f} = \hat{F}/m$ and recall the definitions of ω_n and ζ to obtain a nonhomogeneous linear ODE in ω_n , ζ , $\hat{f}$.
- (7) Write the general solution $u(t)$ as a sum of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.
- (8) Write the general form of the complementary solution $u_c(t)$.
- (9) Find the particular solution $u_p(t)$.
- (10) Use (8), (9) to write the complete solution $u(t)$; discuss the meaning of transient response and steady-state response.
- (11) Find an expression for the steady-state response amplitude $\hat{u}_{ss}$ in terms of ω_n , ζ , $\hat{f}$, and ω .
- (12) Develop the expression for the frequency response function $\text{FRF}(\omega)$ in terms of ω_n , ζ , ω .
- (13) Discuss the variation with frequency of FRF magnitude and phase.

- (14) Give the definition of the magnification factor $M(\omega)$.
- (15) Give the expression for the magnification factor $M(\omega)$ in terms of ω_n and ζ .
- (16) Give the definition of the resonance frequency ω_r .
- (17) Give the expression for the resonance frequency ω_r in terms of ω_n and ζ .
- (18) Give the expression of the peak magnification M_r .
- (19) Give the expression for the FRF phase $\phi(p)$, where $p = \omega/\omega_n$.
- (20) Give the definition of phase resonance and deduce the FRF expression at phase resonance.
- (21) Calculate the magnification factor at phase resonance.
- (22) Calculate the magnitude of the physical response at phase resonance.
- (23) Discuss the difference between true resonance and phase resonance.
- (24) Write and compare the expressions for the resonance frequency ω_r and damped frequency ω_d .
- (25) Establish a formula to calculate the RPM corresponding to a rad/sec frequency ω .

B.7 Calculate Flex-Beam Response to Slightly Unbalanced Electric Motor Excitation

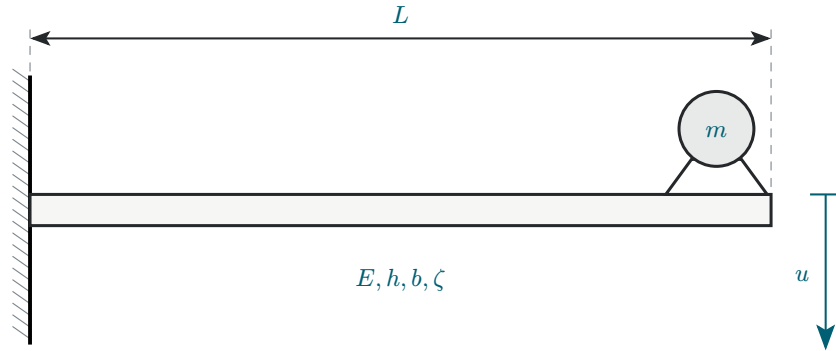


Figure 13: Damped cantilever beam excited by force $F(t)$ from slightly unbalanced electric motor.

Consider a cantilever beam of length $L = 568 \text{ mm}$, elastic modulus $E = 200 \text{ GPa}$, cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$, and damping ratio $\zeta = 14\%$. (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass $m = 0.52 \text{ kg}$ operating at $1,500 \text{ RPM}$ is attached at the beam tip as shown in Figure 13. The electric motor is slightly unbalanced, producing a slight vibratory excitation $F(t) = \hat{F}e^{i\omega t}$ with $\hat{F} = 12 \text{ mN}$. The excitation frequency ω depends on the electric motor speed.

Do the following.

- (1) Display input data.
- (2) Calculate:
 - (a) Cross-section second moment of area I , flexural stiffness EI , and spring constant k .
 - (b) Natural frequency ω_n rad/s, critical damping c_{cr} , physical damping c .
 - (c) Quasi-static deflection u_{qst} corresponding to $\omega_n \rightarrow 0$.
- (3) Calculate the values of:
 - (a) Resonant frequency ω_r and damped frequency ω_d .
 - (b) Peak magnification M_r .
 - (c) Magnification at phase resonance M_{90} .
 - (d) Physical response $\hat{u}_{90}$ at phase resonance and its magnitude $|\hat{u}_{90}|$.
 - (e) Calculate the Hz frequencies f_r , f_n and the corresponding RPM values.
- (4) Assume excitation at phase resonance:
 - (a) Calculate the excitation frequency ω in rad/s and f in Hz.
 - (b) Calculate period of oscillation τ .
 - (c) Plot the time response for $N_{osc} = 10$ oscillations. Hint: plot the real part of the complex displacement $u(t)$.
- (5) Graduate work. Complete items (a)–(d).

- (a) Calculate and display the operational frequency f_0 in Hz and the corresponding ω_0 in rad/sec. Plot the magnitude and phase of the frequency response function FRF vs. excitation f in Hz from quasi-static to f_0 with $N_f = 5000$ points.
- (b) Calculate and display the FRF magnitude at phase-resonance RPM, at double the phase resonance RPM, and at operational RPM. Display the corresponding Hz frequencies. Put the corresponding data tips on the magnitude plot.
- (c) Calculate the vibration magnitude $|u_0|$ when the electric motor runs at its operational RPM.
- (d) Comment on what vibration would be observed as the motor is switched on and accelerates to reach its operational RPM.

B.8 Graduate Work. Calculate Horizontal Flex-Beam Response to Impact

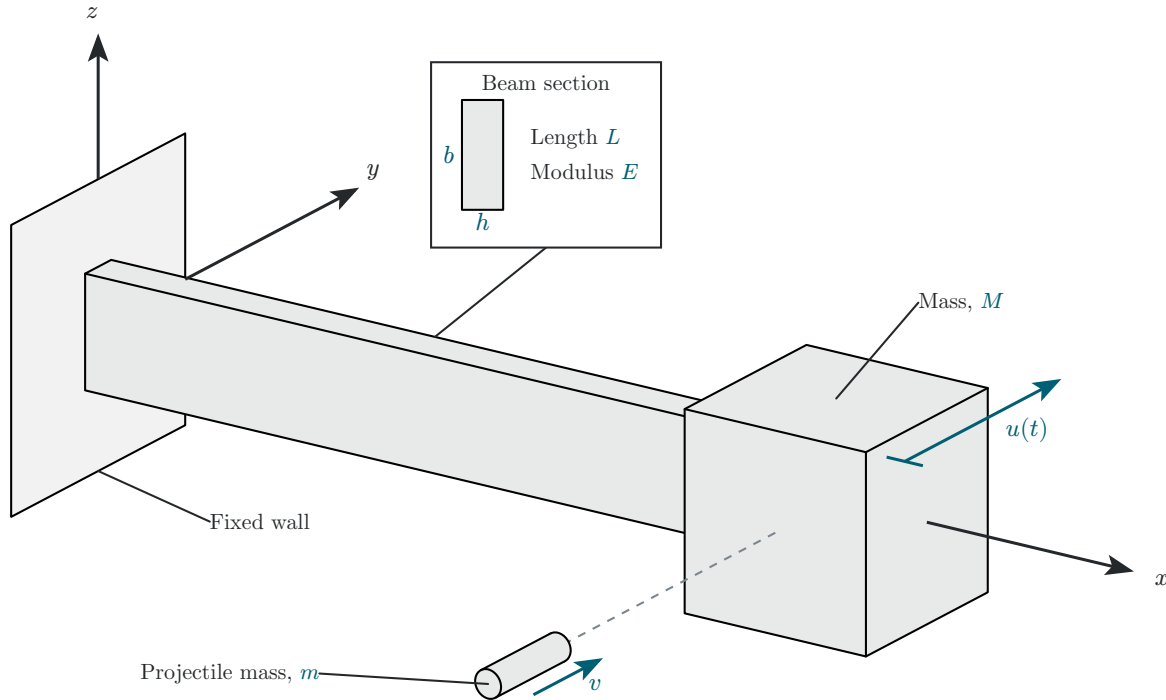


Figure 14: Impact excitation of a horizontal flex-beam vibration system.

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $M = 53 \text{ kg}$ attached at its tip as shown in Figure 14. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The beam is initially at rest. A small projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion $u(t)$ contained in the horizontal plane.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (6) Calculate the rad/sec natural frequency ω_n and period of oscillation τ .
- (7) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.

- (b) The response calculated at item (7).
- (9) Discuss your results.

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Vertical Spring-Mass Vibration System

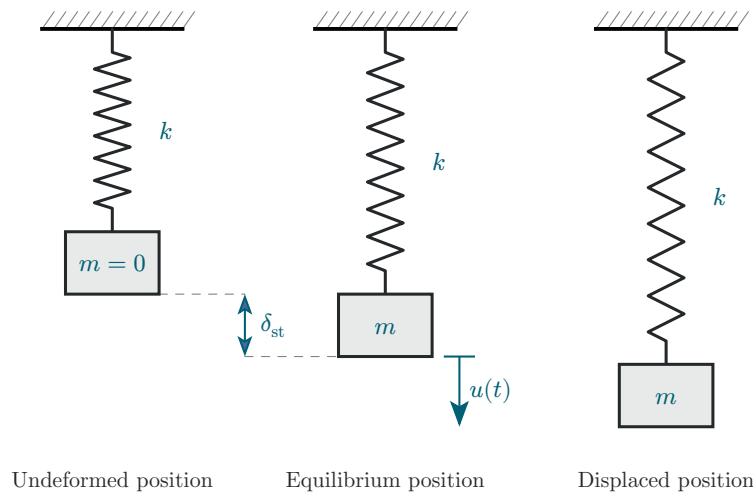


Figure 15: Vertical spring-mass vibration system.

Consider a vertical spring-mass vibration system consisting of a particle of mass m hanging on an elastic spring of stiffness k as shown in Figure 15.

Do the following.

- (1) Assume static equilibrium.
 - (a) Calculate the weight W acting on the mass m .
 - (b) Draw free-body diagram (FBD).
 - (c) Write equilibrium equation.
 - (d) Solve for the static displacement δ_{st} .
- (2) Assume vibration with time-varying vertical displacement $u(t)$.
 - (a) Draw free-body diagram (FBD).
 - (b) Deduce equation of motion (EOM).
 - (c) Compare the EOM deduced here with the EOM derived for horizontal spring-mass vibration system of Problem B.1.
- (3) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

C.2 Analyze Vertical Flex-Beam Vibration

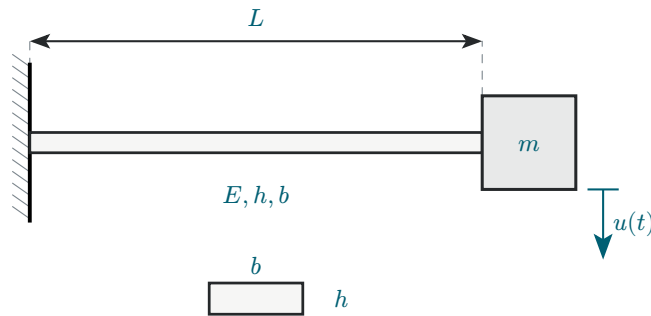


Figure 16: Vertical flex-beam vibrating system.

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 16. The beam cross-section properties are E , h , b .

Do the following.

- (1) Calculate the cross-sectional stiffness EI of the beam for vertical flexural deformation.
- (2) Calculate the spring constant k of the system.
- (3) Assume static equilibrium:
 - (a) Calculate the weight W acting on the mass m .
 - (b) Draw free-body diagram (FBD).
 - (c) Write equilibrium equation.
 - (d) Solve for the static displacement δ_{st} .
- (4) Assume vibration with time-varying vertical displacement $u(t)$ measured from the static equilibrium position:
 - (a) Draw free-body diagram (FBD).
 - (b) Deduce equation of motion (EOM).
 - (c) Compare the EOM deduced here with the EOM derived for horizontal flex-beam vibration system of Problem C.4.
- (5) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

C.3 Analyze Horizontal Flex-Beam Vibration

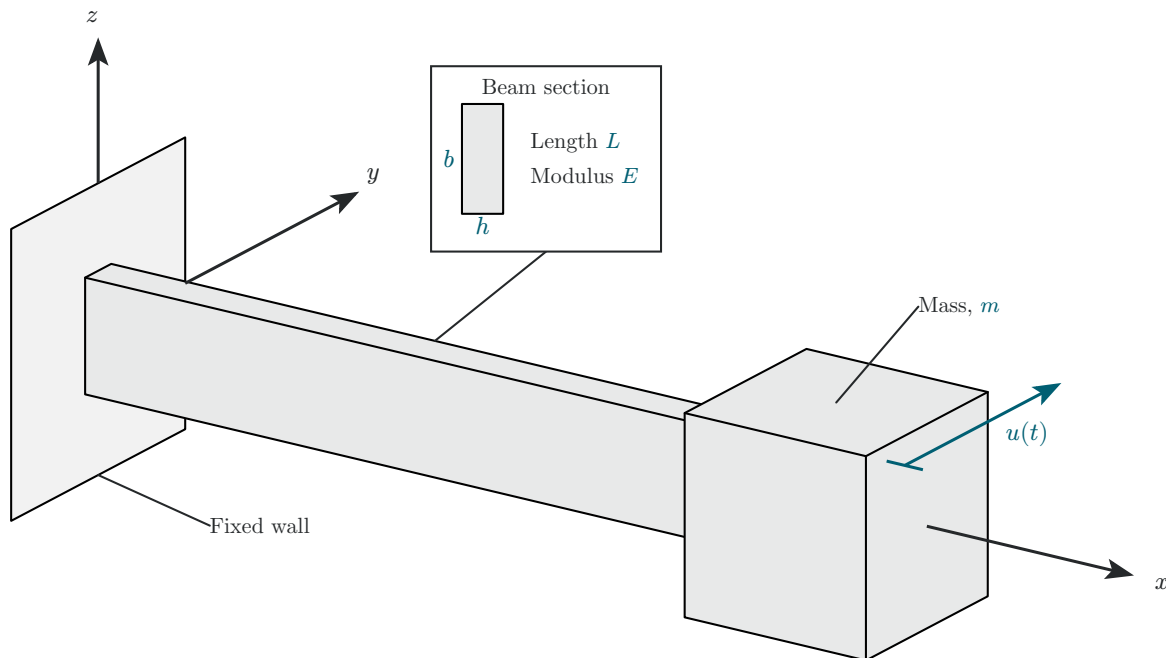


Figure 17: *Horizontal flex-beam vibrating system.*

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 17. The beam cross-section properties are E , h , b .

Do the following.

- (1) Assume vibration with horizontal time-varying displacement $u(t)$ as shown in Figure 17.
- (2) Calculate the cross-sectional stiffness EI of the beam for flexural deformation consistent with $u(t)$.
- (3) Calculate the flex-beam spring constant k .
- (4) Draw free-body diagram (FBD) including all the forces acting on the mass m .
- (5) Deduce equation of motion (EOM).
- (6) Compare the EOM deduced here with the EOM derived in Problem B.1.
- (7) Solve the initial value problem (IVP) and find the response for simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

C.4 Calculate Horizontal Flex-Beam Response to Initial Displacement

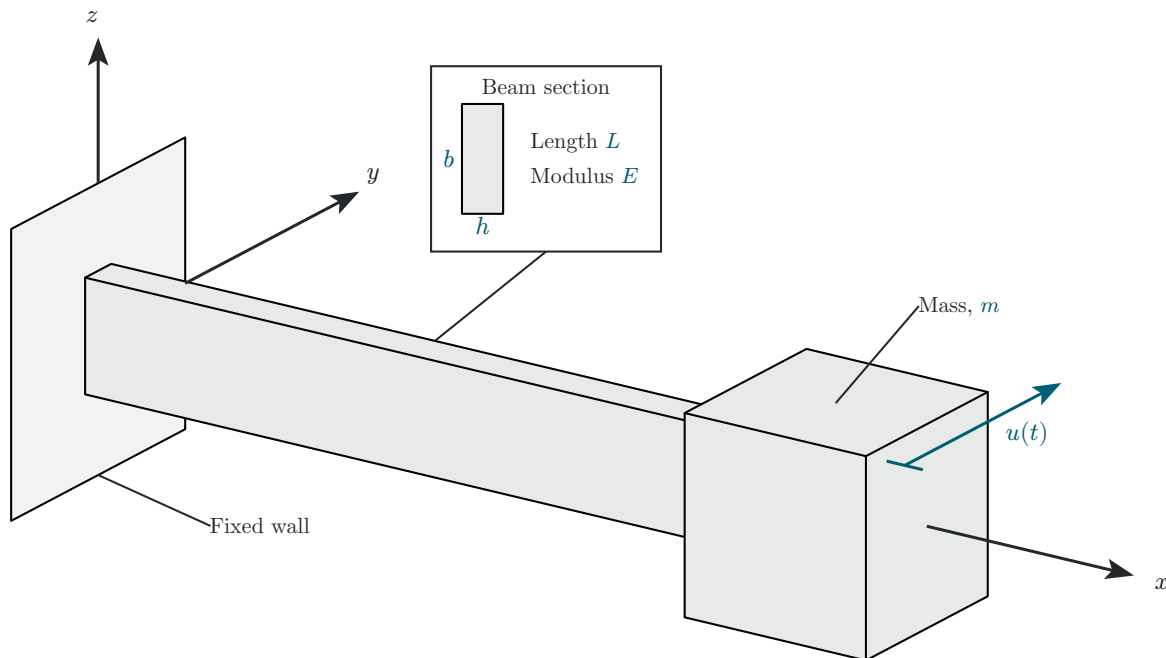


Figure 18: *Flex-beam vibrating system.*

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $m = 5.35 \text{ kg}$ attached at its tip as shown in Figure 18. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The tip mass is given an initial displacement $u_0 = 5.4 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n .
- (6) Calculate the natural frequency in Hz f_n .
- (7) Calculate the period of oscillation in sec τ .
- (8) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (9) Plot the response of the system for $N_{\text{osc}} = 25$ oscillations and use a data tip to find $u(t_1)$ at t_1 .
- (10) Discuss your results.

C.5 Analyze Spring-Mass Forced Vibration

Consider a spring-mass vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 19.

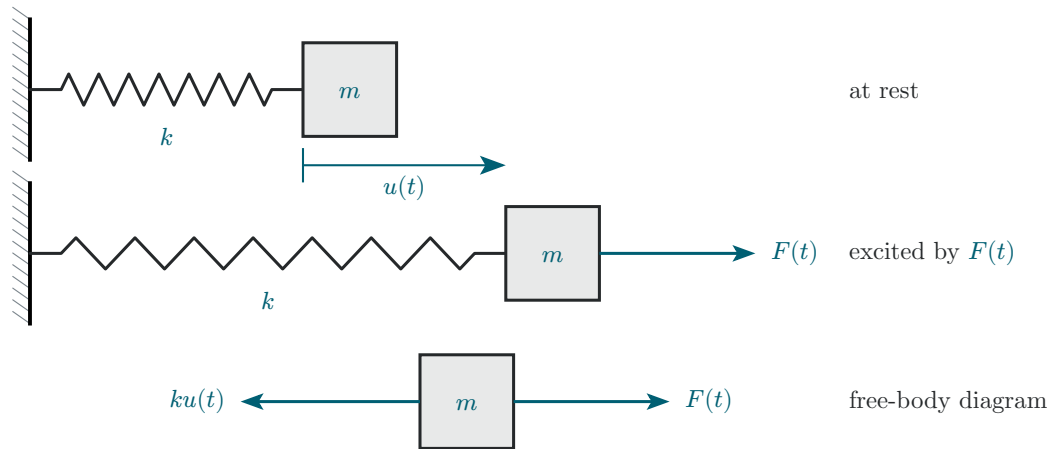


Figure 19: Spring-mass system excited by force $F(t)$.

General Analysis of Spring-Mass Forced Vibration

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) and express it as a nonhomogeneous ODE in terms of m , k , $F(t)$.
- (5) Write the ODE general solution as the superposition of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.
- (6) Recall the complementary solution $u_c(t)$ as the solution of the homogeneous equation derived in Problem B.1 and write it in terms of trigonometric functions multiplied by constants A and B .

Spring-Mass Response to a Suddenly Applied Load

- (7) Assume the loading $F(t)$ is a suddenly applied load of magnitude F_0 .
- (8) Find the particular solution for this case of a suddenly applied load.
- (9) Write the complete solution as the sum of $u_c(t)$ from (6) and $u_p(t)$ from (8).
- (10) Assume “at rest” initial conditions to find the constants A and B and write the complete solution $u(t)$ as a function of F_0 , k , and $\cos \omega_n t$.
- (11) Use (10) to find the largest displacement $u_{\max}$ and compare it with the static displacement u_{st} .

C.6 Calculate Cantilever Response to Suddenly Applied Weight

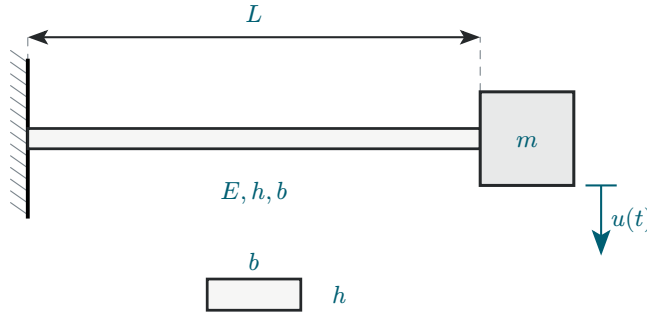


Figure 20: Cantilever beam with tip mass.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length $L = 568 \text{ mm}$ and a mass $m = 5.35 \text{ kg}$ that has to be attached to its tip as shown in Figure 20. The beam is made of steel with elastic modulus $E = 200 \text{ GPa}$, yield stress $Y = 710 \text{ MPa}$, and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$.

Do the following.

- (1) Display input data.
- (2) Calculate the cross-section second moment of area I , the flexural stiffness EI , the spring constant k , the natural frequency ω_n rad/s, f_n Hz, and period of oscillation τ of the system.
- (3) Calculate the weight F_0 of the mass m .
- (4) Static case. Assume the mass is attached to the undeformed beam and then released gradually without inducing vibration. Calculate the static displacement u_{st} .
- (5) Dynamic case. Assume the mass is attached to the undeformed beam and then released suddenly, inducing vibration.
 - (a) Calculate the maximum dynamic displacement u_{dyn} .
 - (b) Plot the response $u(t)$ measured from the undeformed position for $N_{\text{osc}} = 10$ oscillations.
 - (c) Place a data tip on the plot to identify the maximum displacement and compare it with the result of item (5)(a).
- (6) Graduate work. Calculate the maximum stress at the root of the beam and the safety factor $\text{SF} = Y/\sigma$, and comment on your results for the following two cases.
 - (a) Static-case stress σ_{st} .
 - (b) Dynamic-case stress σ_{dyn} .
- (7) Discuss your results.

C.7 Analyze Flex-Beam Pendulum

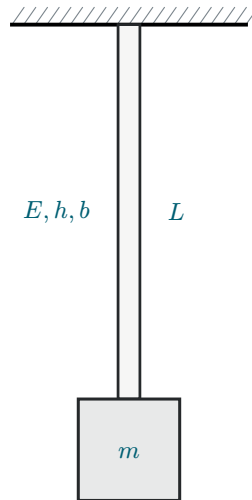


Figure 21: *Flex-beam pendulum.*

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 21. The beam is fixed into the ceiling and is hanging downward along the z axis. The mass may oscillate sideways in either x or y directions. The beam cross-section properties are E, h, b .

Do the following.

- (1) Calculate the minor and major second moments of area I_1 and I_2 .
- (2) Assume a horizontal displacement $u(t)$ which produces flexure about the minor I_1 .
- (3) Draw free-body diagram (FBD).
- (4) Write Newton's law of motion (NLM) equation.
- (5) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (6) Define the natural frequency $\omega_{n,\text{FBP}}$ of the flex-beam pendulum.
- (7) Deduce the governing equation.
- (8) Compare your results with those of Problem B.1 and write the general solution.
- (9) Solve the initial value problem (IVP), i.e., deduce the response for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.
- (10) Graduate work. Explain how the solution would change if flexure were about I_2 .

C.8 Calculate Flex-Beam Pendulum Response to Impact

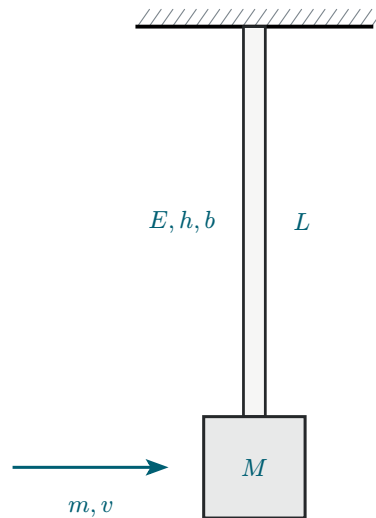


Figure 22: Impact excitation of a vertical flex-beam pendulum.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length $L = 568 \text{ mm}$ with a mass $M = 42 \text{ kg}$ attached at its end, as shown in Figure 22. The beam is steel with $E = 200 \text{ GPa}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$. The beam is initially at rest. A projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. The projectile orientation causes the beam to flex about its minor cross-sectional axis. The impact imparts an initial velocity to the system, which then oscillates.

Do the following.

- (1) Display input data.
- (2) Calculate the minor second moment of area I of the beam.
- (3) Calculate the minor cross-sectional stiffness EI .
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as resulting from the interaction between the projectile and the large mass.
- (6) Calculate the rad/s natural frequency $\omega_{n,\text{FBP}}$ and period of oscillation τ_{FBP} of the flex-beam pendulum with projectile embedded into it.
- (7) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (7).
- (9) Compare your results with those of Problem A.3.

C.9 Complex Exponentials and Trigonometric Functions

- (1) Find the solutions of $z^4 = 1$ (the fourth roots of unity) and express them as:
 - (a) Complex numbers.
 - (b) Complex exponentials.
- (2) Write the complex number $e^{i\omega t}$ as a sum of trigonometric functions (hint, Euler's formula).
- (3) Calculate the phase difference ϕ in radians and degrees between the following signals:
 - (a) Compare $x_1(t) = \sin(\omega t)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.
 - (b) Compare $x_1(t) = \sin(\omega t + \pi/3)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.
 - (c) Compare $x_1(t) = \sin(\omega t + 45^\circ)$ and $x_2(t) = \sin(\omega t + 115^\circ)$.
- (4) What is the phase in radians and degrees of the following complex numbers:
 - (a) $e^{i\pi/2}$.
 - (b) $e^{i3\pi/2}$.
 - (c) $e^{i\pi/4}$.
 - (d) $e^{-i\pi/2}$.
 - (e) $e^{-i\pi/4}$.
- (5) Draw the locations on the unit circle of the complex numbers listed in item (4).
- (6) Calculate the magnitude and phase in degrees of the complex number $Z = 4 + 5i$.